CS2201 Computer Organization & Architecture

L4: Multiplication of Binary Numbers



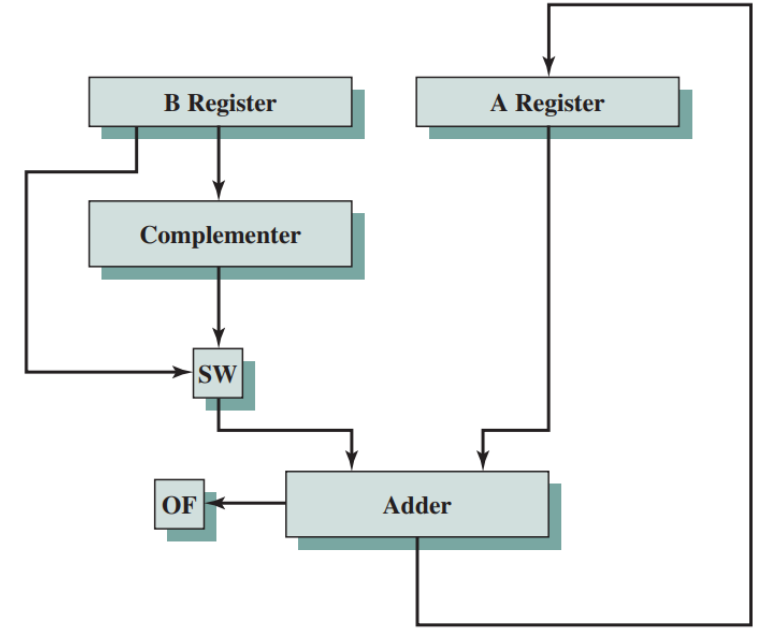
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Addition & Subtraction Circuits



OF = Overflow bit
SW = Switch (select addition or subtraction)

- Addition and subtraction is done in same circuit with a mode switch.
- Subtraction is achieved using addition.
- To subtract one number (subtrahend) from another (minuend), take the 2' complement (negation) of the subtrahend and add it to the minuend.

Binary Multiplication

Multiplicand	:	4	[0100]
Multiplier	: X	3	[0011]
Product	:	12	[01100]

Binary Multiplication

A	B	A X B
0	0	0
0	1	0
1	0	0
1	1	1

Multiplication of Unsigned Integers

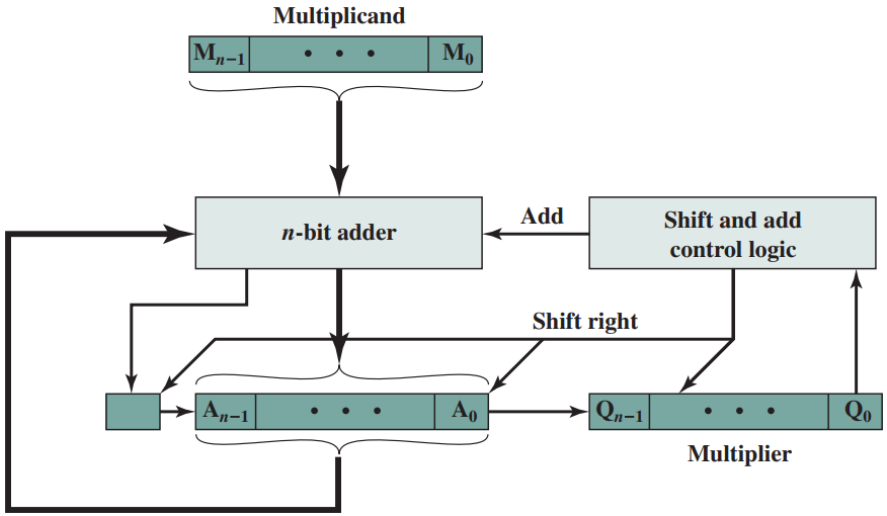
```
23      10111     Multiplicand
19    × 10011     Multiplier
        10111
       10111
      00000   +
     00000
    10111
437 110110101     Product
```

- Multiplication involves the generation of partial products (PP).
- One PP for each digit in the multiplier.
- PPs are then added for the final product.

- When the multiplier bit is 0, PP is 0.
- When the multiplier is 1, PP is the multiplicand.
- The multiplication of two n-bit binary integers, product up to 2n bits

  Eg., 11 x 11 = 1001.

Multiplication of Unsigned Integers

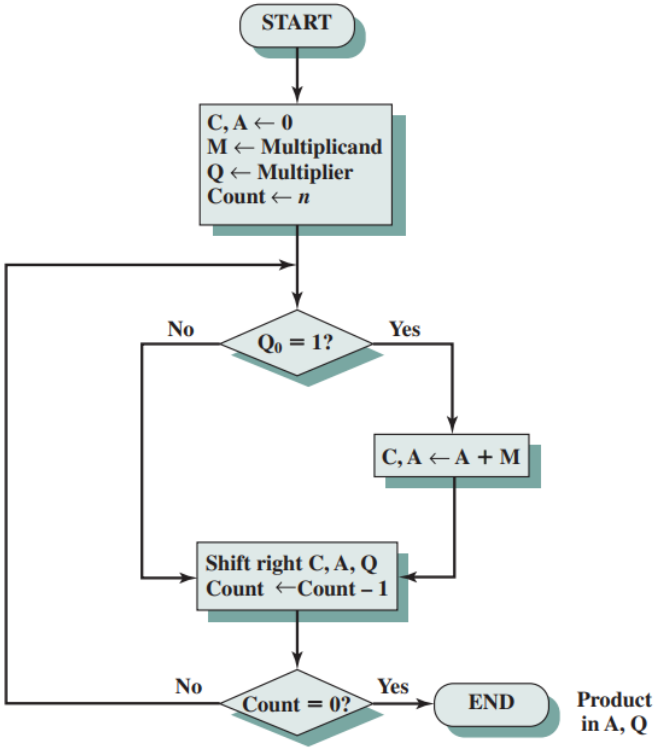


- Perform a running addition on the partial products rather than waiting until the end.
- 1 on the multiplier → add + shift
- 0 on the multiplier → shift

- Eliminates the need for storage of all the partial products
- Fewer registers are needed.

Multiplication of Unsigned Integers

C	A	Q	M		
0	0000	1101	1011	Initial values	
0	1011	1101	1011	Add	First cycle
0	0101	1110	1011	Shift	
0	0010	1111	1011	Shift	Second cycle
0	1101	1111	1011	Add	Third cycle
0	0110	1111	1011	Shift	
1	0001	1111	1011	Add	Fourth cycle
0	1000	1111	1011	Shift	



11 x 13 = 143; i.e., 1000 1111

Booth's Multiplication

- Three symbols to represent numbers: 1, 0, -1
- Bit transitions show Booth's encoding

Q_{n-1}, Q_n

0 → 0: 0

0 → 1: -1

1 → 1: 0

1 → 0: 1



$$M \times \boxed{0}\boxed{0}\boxed{1}\boxed{1}\boxed{1}\boxed{1}\boxed{1}\boxed{0} = M \times (2^5 + 2^4 + 2^3 + 2^2 + 2^1) = M \times 62$$

$$M \times \boxed{0}\boxed{1}\boxed{0}\boxed{0}\boxed{0}\boxed{0}\boxed{-1}\boxed{0} = M \times (2^6 - 2^1) = M \times 62$$



$$M \times \boxed{0}\boxed{0}\boxed{1}\boxed{1}\boxed{1}\boxed{0}\boxed{1}\boxed{0} = M \times (2^5 + 2^4 + 2^3 + 2^1) = M \times 58$$

$$M \times \boxed{0}\boxed{1}\boxed{0}\boxed{0}\boxed{-1}\boxed{1}\boxed{-1}\boxed{0} = M \times (2^6 - 2^3 + 2^2 - 2^1) = M \times 58$$

Booth's Multiplication

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Q_{n-1}, Q_n

0 → 0: 0

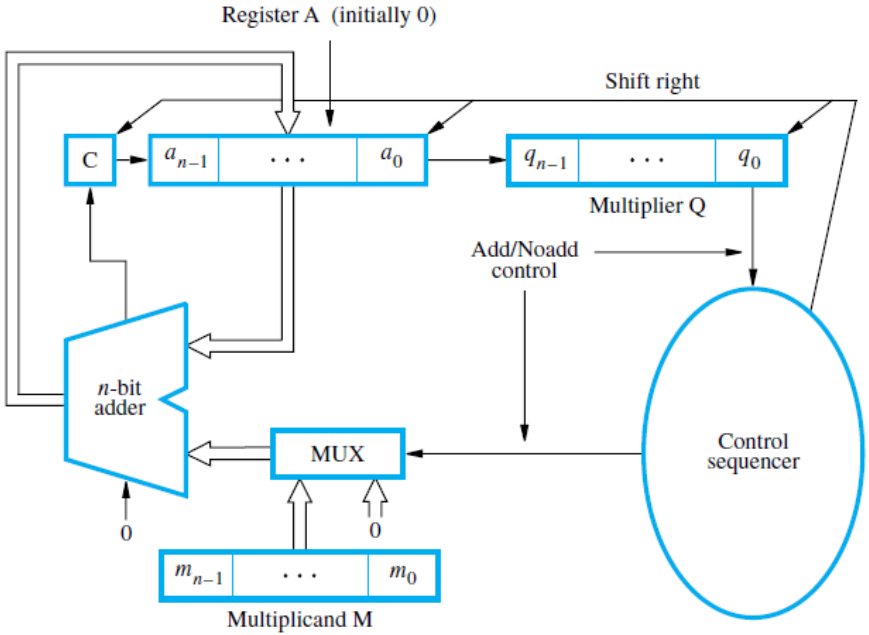
0 → 1: -1

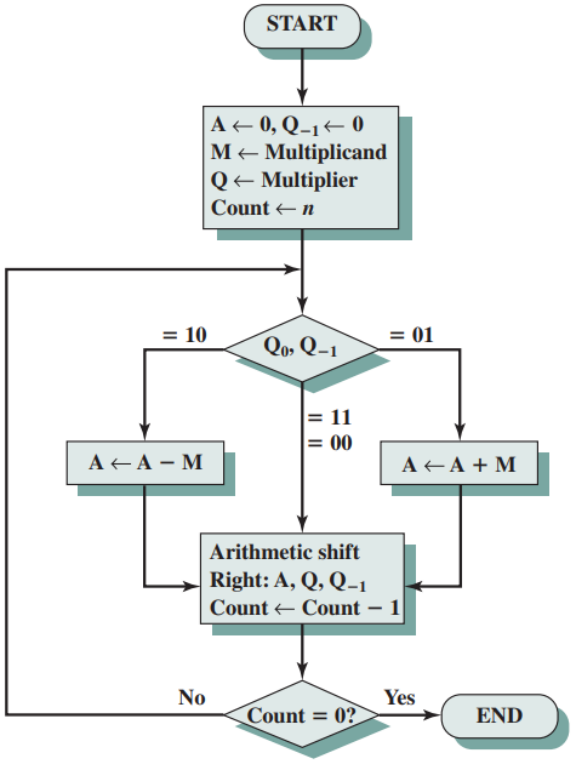
1 → 1: 0

1 → 0: 1

- -1 in 8 bits
  - 11111111 (2's C)
  - 0000000-1 (Booth's encoding)
- 14 in 8 bits
  - 00001110 (2's C)
  - 000100-10 (Booth's encoding)

Booth's Multiplication





Q_{-1}, Q_0

0 → 0: **0**

0 → 1: **-1**

1 → 1: **0**

1 → 0: **1**

Booth's Algorithm Example

2 x -3 : 0010 x 1101

Q =1101

M =0010

-M=1110

Iteration	Implementation		
	Step	M / -M	A Q Q_{-1}
0	initial values	0010 / 1110	0000 1101 0
1			
2			
3			
4			

Booth's Algorithm Example

2 x -3 : 0010 x 1101

Q =1101

M =0010

-M=1110

Iteration	Implementation		
	Step	M / -M	A Q Q_{-1}
0	initial values	0010 / 1110	0000 1101 0
1	10 → A = A - M	0010 / 1110	1110 1101 0
	shift right		1111 0110 1
2			
3			
4			

Booth's Algorithm Example

2 x -3 : 0010 x 1101

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Iteration	Implementation		
	Step	M / -M	A Q Q_{-1}
0	initial values	0010 / 1110	0000 1101 0
1	10 → A = A - M	0010 / 1110	1110 1101 0
	shift right		1111 0110 1
2	01 → A = A + M	0010 / 1110	0001 0110 1
	shift right		0000 1011 0
3			
4			

Booth's Algorithm Example

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	Step	M / -M	A Q Q_{-1}
0	initial values	0010 / 1110	0000 1101 0
1	10 → A = A - M	0010 / 1110	1110 1101 0
	shift right		1111 0110 1
2	01 → A = A + M	0010 / 1110	0001 0110 1
	shift right		0000 1011 0
3	10 → A = A - M	0010 / 1110	1110 1011 0
	shift right		1111 0101 1
4			

Booth's Algorithm Example

2 x -3 : 0010 x 1101

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M =0010

-M=1110

Iteration	Implementation		
	Step	M / -M	A Q Q_{-1}
0	initial values	0010 / 1110	0000 1101 0
1	10 → A = A - M	0010 / 1110	1110 1101 0
	shift right		1111 0110 1
2	01 → A = A + M	0010 / 1110	0001 0110 1
	shift right		0000 1011 0
3	10 → A = A - M	0010 / 1110	1110 1011 0
	shift right		1111 0101 1
4	11 → NO OP	0010 / 1110	1111 0101 1
	shift right		1111 1010 1

1111 1010 (-ve)
→ 0000 0101 +
1

0000 0110 (-6)

Booth's Algorithm Example

7 x 3: 0111 x 0011

Q =0011

M =0111

-M=1001

Iteration	Implementation		
	Step	M / -M	A Q Q_{-1}
0	initial values	0111 / 1001	0000 0011 0

Booth's Algorithm Example

7 x 3: 0111 x 0011

Q =0011

M =0111

-M=1001

Iteration	Implementation		
	Step	M / -M	A Q Q_{-1}
0	initial values	0111 / 1001	0000 0011 0
1	10 → A = A - M	0111 / 1001	1001 0011 0
	shift right		1100 1001 1
2			
3			
4			

Booth's Algorithm Example

7 x 3: 0111 x 0011

Q =0011

M =0111

-M=1001

Iteration	Implementation		
	Step	M / -M	A Q Q_{-1}
0	initial values	0111 / 1001	0000 0011 0
1	10 → A = A - M	0111 / 1001	1001 0011 0
	shift right		1100 1001 1
2	11 → NO OP	0111 / 1001	1100 1001 1
	shift right		1110 0100 1
3			
4			

Booth's Algorithm Example

7 x 3: 0111 x 0011

Q =0011

M =0111

-M=1001

Iteration	Implementation		
	Step	M / -M	A Q Q_{-1}
0	initial values	0111 / 1001	0000 0011 0
1	10 → A = A - M	0111 / 1001	1001 0011 0
	shift right		1100 1001 1
2	11 → NO OP	0111 / 1001	1100 1001 1
	shift right		1110 0100 1
3	01 → A = A + M	0111 / 1001	0101 0100 1
	shift right		0010 1010 0
4			

Booth's Algorithm Example

7 x 3: 0111 x 0011

Q =0011

M =0111

-M=1001

0001 0101(+ve)
→ 21

Iteration	Implementation		
	Step	M / -M	A Q Q_{-1}
0	initial values	0111 / 1001	0000 0011 0
1	10 → A = A - M	0111 / 1001	1001 0011 0
	shift right		1100 1001 1
2	11 → NO OP	0111 / 1001	1100 1001 1
	shift right		1110 0100 1
3	01 → A = A + M	0111 / 1001	0101 0100 1
	shift right		0010 1010 0
4	00 → NO OP	0111 / 1001	0010 1010 0
	shift right		0001 0101 0

Booth's Algorithm Example

10 x 15: 01010 x 01111

Q =01111

M =01010

-M=10110

10 x 15: 01010 x 01111

= 10 X (16-1)

Iteration	Implementation				
	Step	M / -M	A	Q	Q_{-1}
0	initial values	01010 / 10110	00000	01111	0

Booth's Algorithm Example

10 x 15: 01010 x 01111

Q =01111

M =01010

-M=10110

Iteration	Implementation		
	Step	M / -M	A Q Q_{-1}
0	initial values	01010 / 10110	00000 01111 0
1	10 → A=A-M, SHIFT		
2	11→NO OP, SHIFT		
3	11→NO OP, SHIFT		
4	11→NO OP, SHIFT		
5	01→A=A+M, SHIFT		

10 x 15: 01010 x 01111

= 10 X (16-1)

0110101010101010 X 0111100111111111001 = ?

0110101010101010 X 1000-1010000000-101-1

(6 additions and 18 shifts)



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